

Exercise 7: Implementing Constructor and Setter Injection

BookRepository.java:

```
package com.library.repository;

public class BookRepository {
    public void executeTask() {
        System.out.println("BookRepository: Task executed.");
    }
}
```

BookService.java:

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    public BookService() {}

    public BookService(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
        System.out.println("BookService: Constructor injection triggered.");
    }

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
        System.out.println("BookService: Setter injection triggered.");
    }

    public void listBooks() {
        bookRepository.executeTask();
    }
}
```

applicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
```

```

    http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository" class="com.library.repository.BookRepository" />

    <bean id="bookService" class="com.library.service.BookService">
        <constructor-arg ref="bookRepository" />
        <property name="bookRepository" ref="bookRepository" />
    </bean>

</beans>

```

LibraryApplication.java:

```

package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class LibraryApplication {
    public static void main(String[] args) {
        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");
        BookService bookService = (BookService) context.getBean("bookService");
        bookService.listBooks();
    }
}

```

Output:

```

adityahargunani@Adityas-MacBook librarymanagement % mvn clean compile
[INFO] Building librarymanagement 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.1.0:clean (default-clean) @ librarymanagement ---
[INFO] Deleting /Users/adityahargunani/Downloads/Exercise 7- Implementing Constructor and Setter Injection/librarymanagement/target
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ librarymanagement ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ librarymanagement ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 4 source files to /Users/adityahargunani/Downloads/Exercise 7- Implementing Constructor and Setter Injection/librarymanagement/target/classes
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.685 s
[INFO] Finished at: 2026-06-25T13:41:42+05:30
[INFO]
adityahargunani@Adityas-MacBook librarymanagement % mvn exec:java -Dexec.mainClass="com.library.LibraryApplication"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example.librarymanagement >-----
[INFO] Building librarymanagement 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ librarymanagement ---
BookService: Constructor injection triggered.
BookService: Setter injection triggered.
BookRepository: Task executed.
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.302 s
[INFO] Finished at: 2026-06-25T13:41:48+05:30
[INFO]
adityahargunani@Adityas-MacBook librarymanagement %

```

